

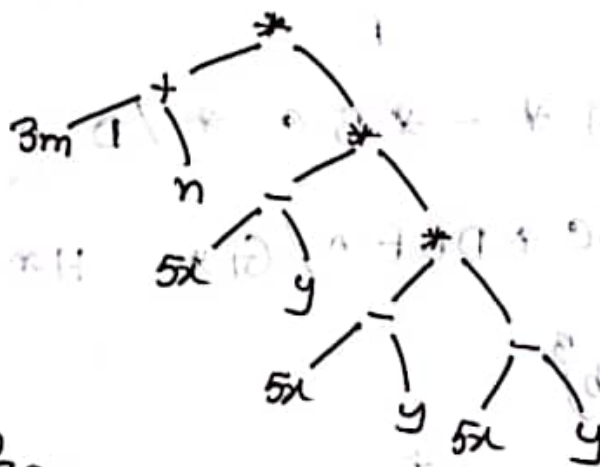
Autumn-22

(4a)

Consider the algebraic expression $E = (3m+n)(5x-y)^3$

(a) Draw the tree T which corresponds to the expression E

$$\begin{aligned} E &= (3m+n) * (5x-y)^3 \\ &= (3m+n) * (5x-y) * (5x-y) * (5x-y) \end{aligned}$$



(b) Find Preorder and postorder

Preorder:

(Root L, R)

* + 3m n * - 5x y *
- 5x y - 5x y

Postorder: $3m \quad n + 5x \quad y - 5x \quad y - 5x \quad y - \dots$
 (L, R, Root) *

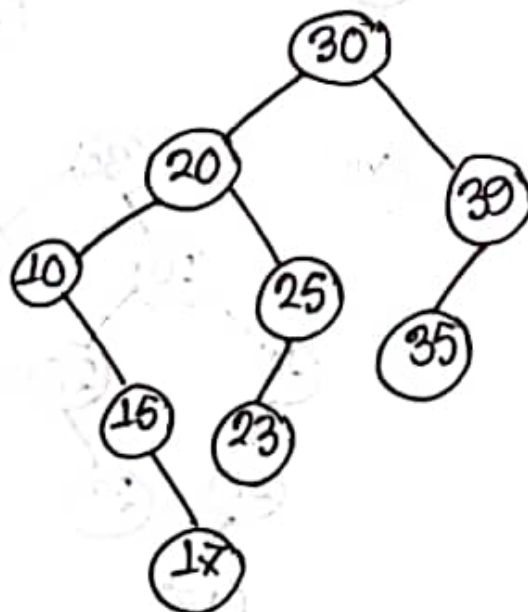
(46)

The Preorder Traversal sequence of a binary search tree is 30, 20, 10, 15, 25, 23, 39, 35, 17

Draw the tree show each step.

Preorder: 30, 20, 10, 15, 25, ~~23~~, ~~39~~, ~~35~~, ~~17~~ [Root, L, R]

Inorder: 10, 15, 17, 20, 23, 25, 30, 35, 39 [L, Root, R]



(4c)

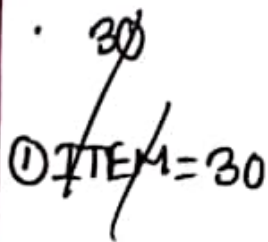
What do you mean by max heap? Build a max heap from the following list of numbers:

30, 39, 29, 27, 37, 52, 63, 44

⇒ Heap or maxheap is a specialized tree-based data structure which is essential on almost complete tree that satisfy the heap property.

Maxheap → The value of N is greater than or equal to children.

Minheap → The value of N is less than children.



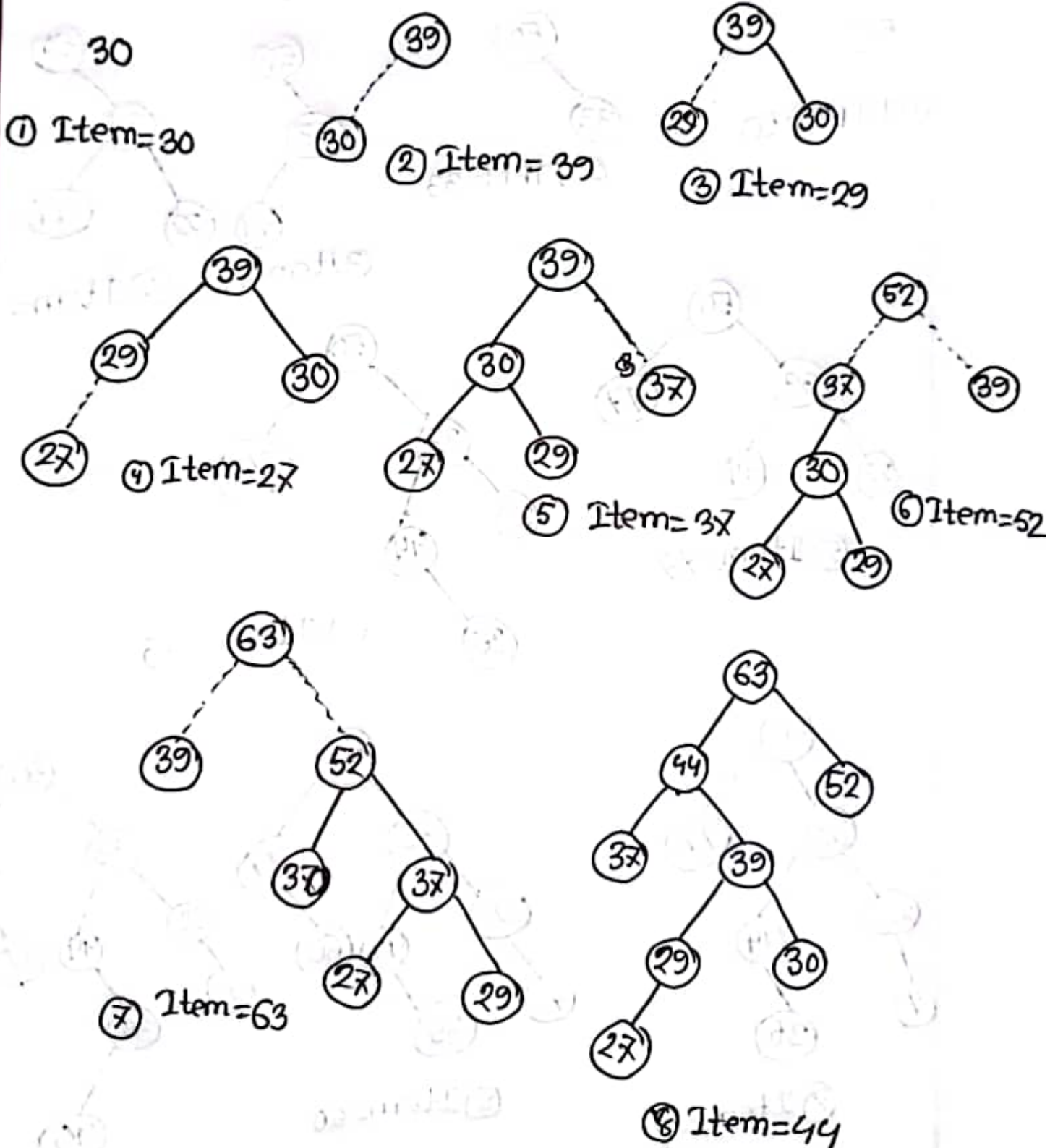
30

① Item = 30



(40)

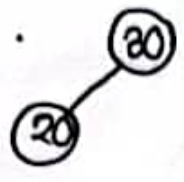
30, 39, 29, 27, 37, 52, 63, 44



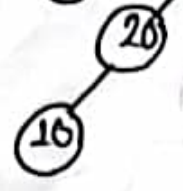
Show steps:

(i) 30

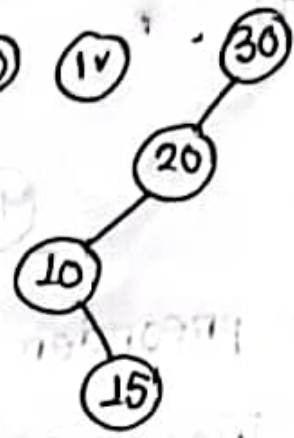
(ii)



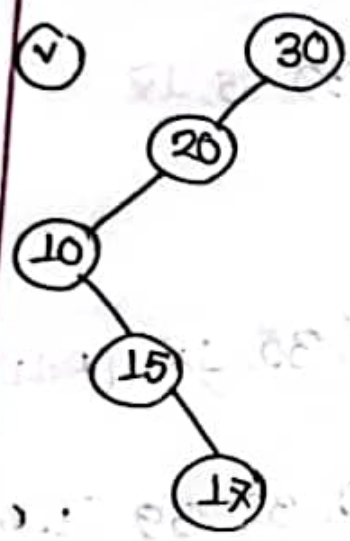
(iii)



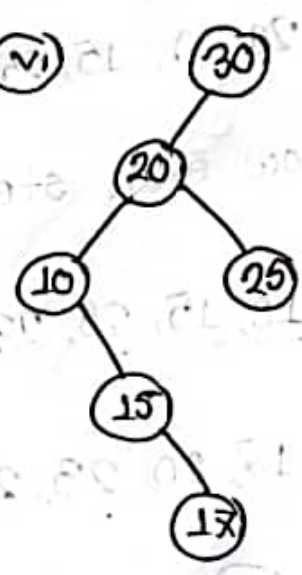
(iv)



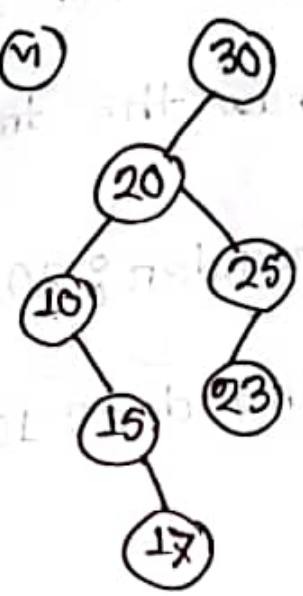
(v)



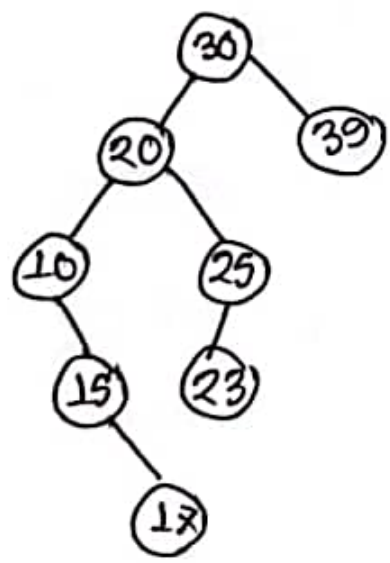
(vi)



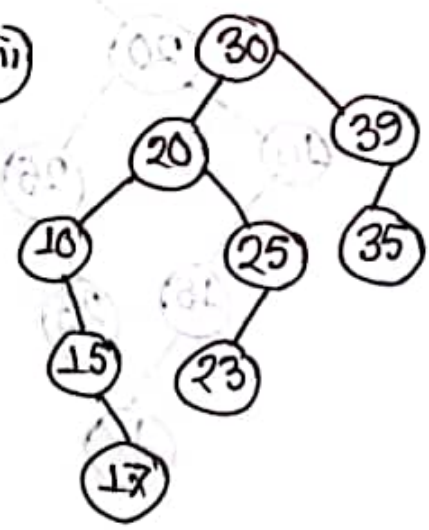
(vii)



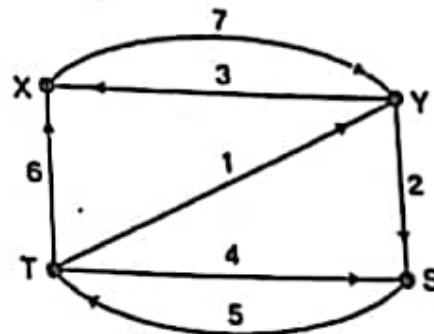
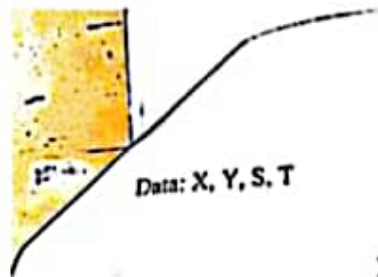
(viii)



(ix)



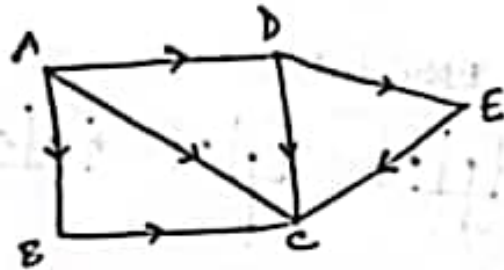
Autumn 22



Find the weighted matrix W of the weighted graph. Then find Q_0 and Q_1 from W using Warshall's algorithm.

In graph, An adjacency list is a collection of unordered list used to represent a finite graph. An adjacency list represents a graph as an array of linked lists.

e) Let us consider the graph drawn below :



Adjacency List of this graph :

A : B, C, D

B : C

C :

D : C, E

E : C

The link representation of this graph will contain two lists, → NODE list and EDGE list.

Each element in the NODE list will be a record of the form :

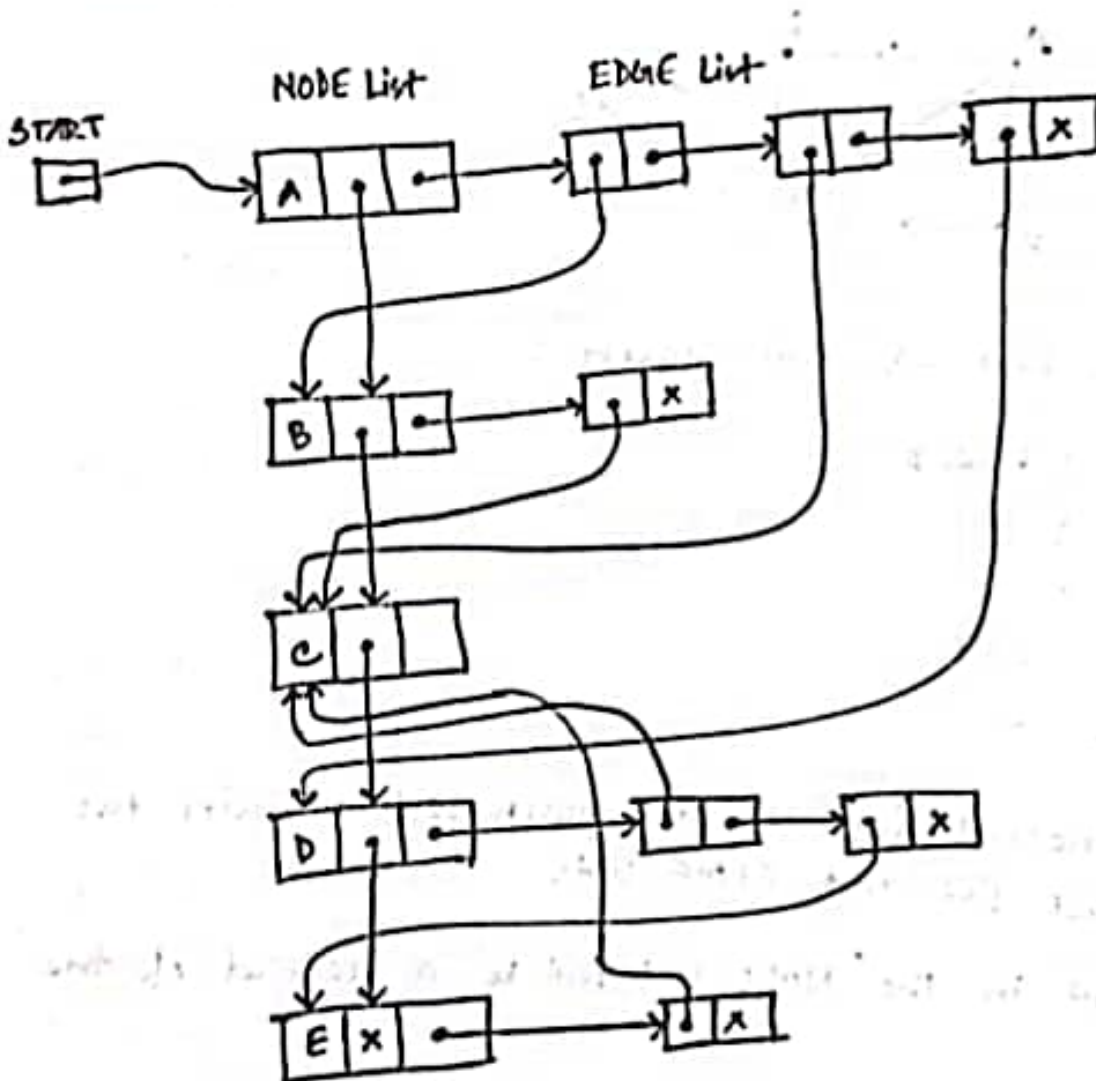
NODE	NEXT	ADJ	
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For more information you can see that SCHAUM'S book page -8.12-8.13

Each element in the EDGE list will correspond to an edge of graph and will be a record of the form:

DEST	LNK	
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The linked representation of the graph given above is as follows:



(b)

The weighted matrix w of the weighted graph is given below:

	x	y	s	T
x	0	7	0	0
y	3	0	2	0
s	0	0	0	5
T	6	1	4	0

$$w = \begin{pmatrix} 0 & 7 & 0 & 0 \\ 3 & 0 & 2 & 0 \\ 0 & 0 & 0 & 5 \\ 6 & 1 & 4 & 0 \end{pmatrix}$$

$$g_0 = \begin{bmatrix} \infty & 7 & \infty & \infty \\ 3 & \infty & 2 & \infty \\ \infty & \infty & \infty & 5 \\ 6 & 1 & 4 & \infty \end{bmatrix}$$

$$g_1[2,2] = \min \{ g_1[2,2], g[2,1] + g[1,2] \}$$

$$= 10$$

$$g_1[2,3] = 2$$

$$g_1[2,4] = \infty$$

$$g_1[3,2] = \infty$$

$$g_1[3,3] = \infty$$

$$g_1[3,4] = 5$$

$$g_1[4,2] = 1$$

$$g_1[4,3] = 4$$

$$Q_1[4,4] = \infty$$

$$Q_1 = \begin{bmatrix} \infty & 7 & \infty & \infty \\ 3 & 10 & 2 & \infty \\ \infty & \infty & \infty & 5 \\ 6 & 1 & 4 & \infty \end{bmatrix}$$